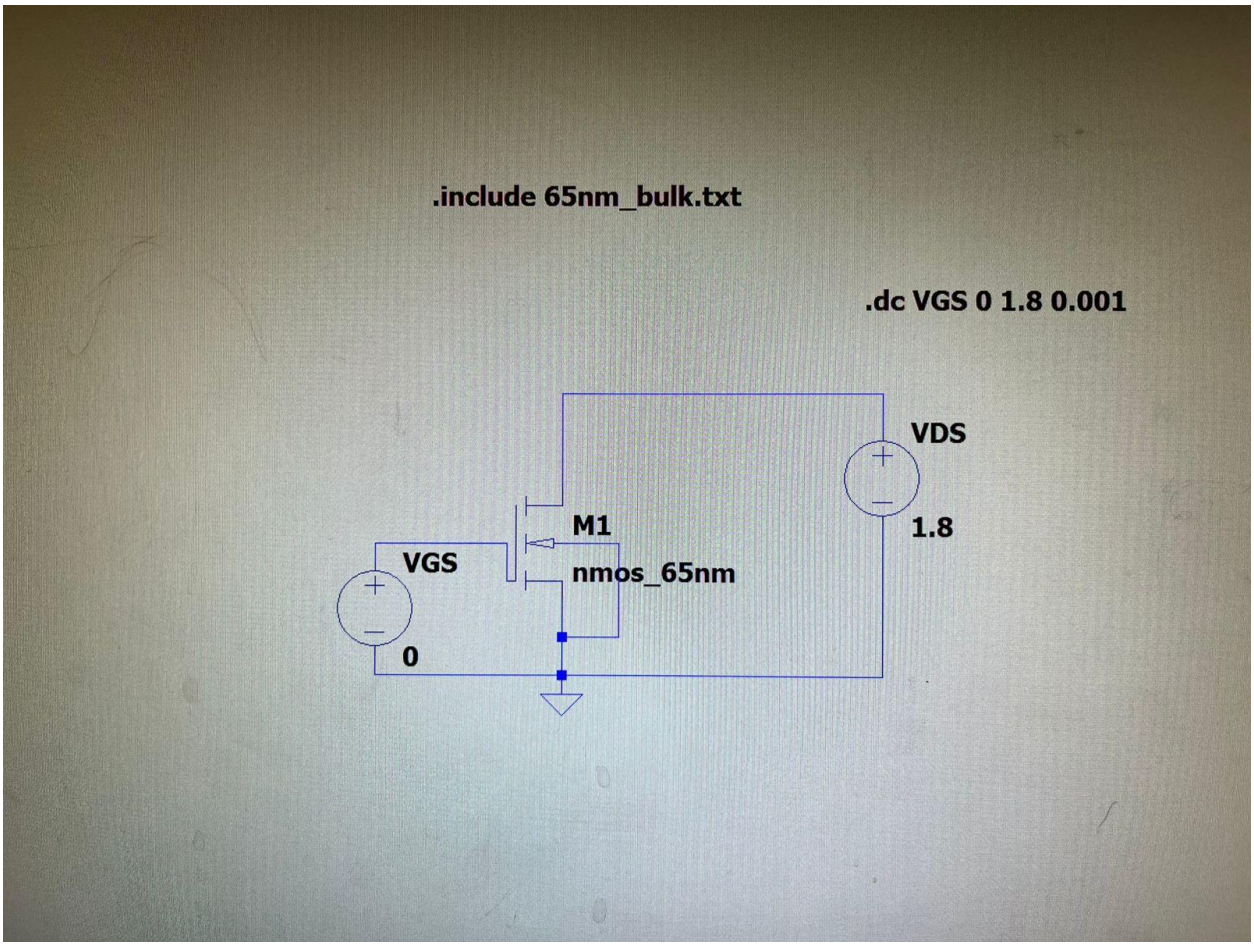
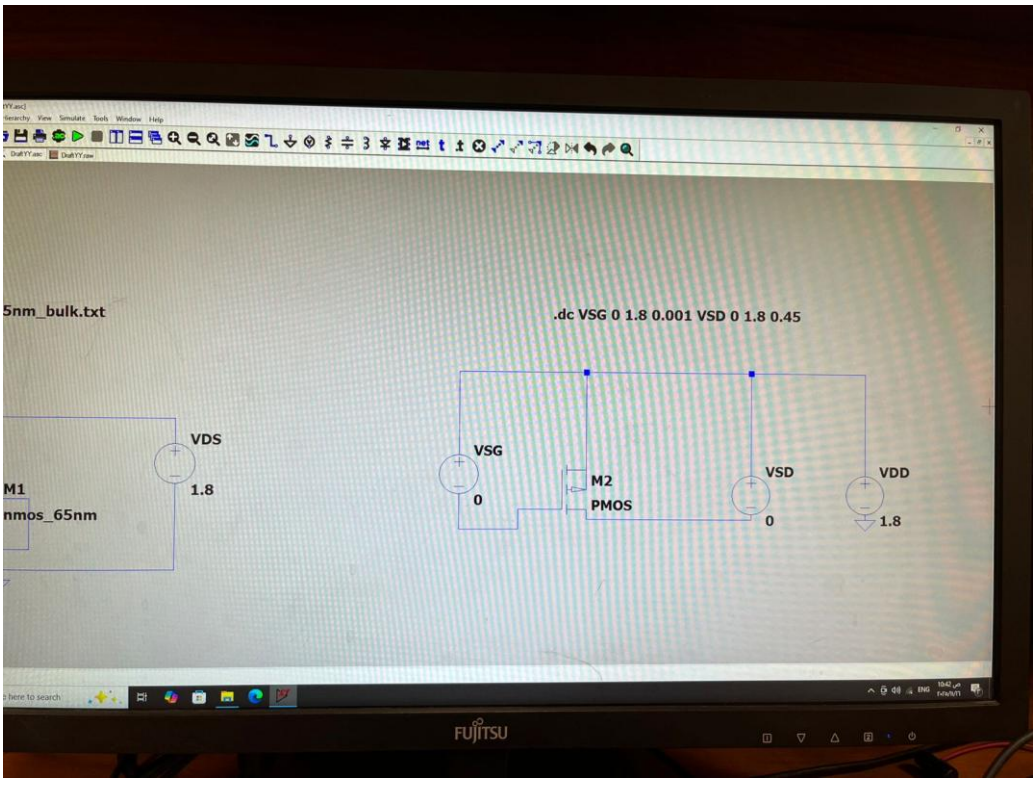
Lab 3 Digital Circuits

Name	Section	Attendance slot	Group No.
Amr Hatem Sabry Elmancy	1	Sunday (from 10:00 AM to 12:00 PM) second rotation	14

Name	Section	Attendance slot	Group No.
Abdelrahman Sherif Medhat	1	Sunday (from 10:00 AM to 12:00 PM) second rotation	14

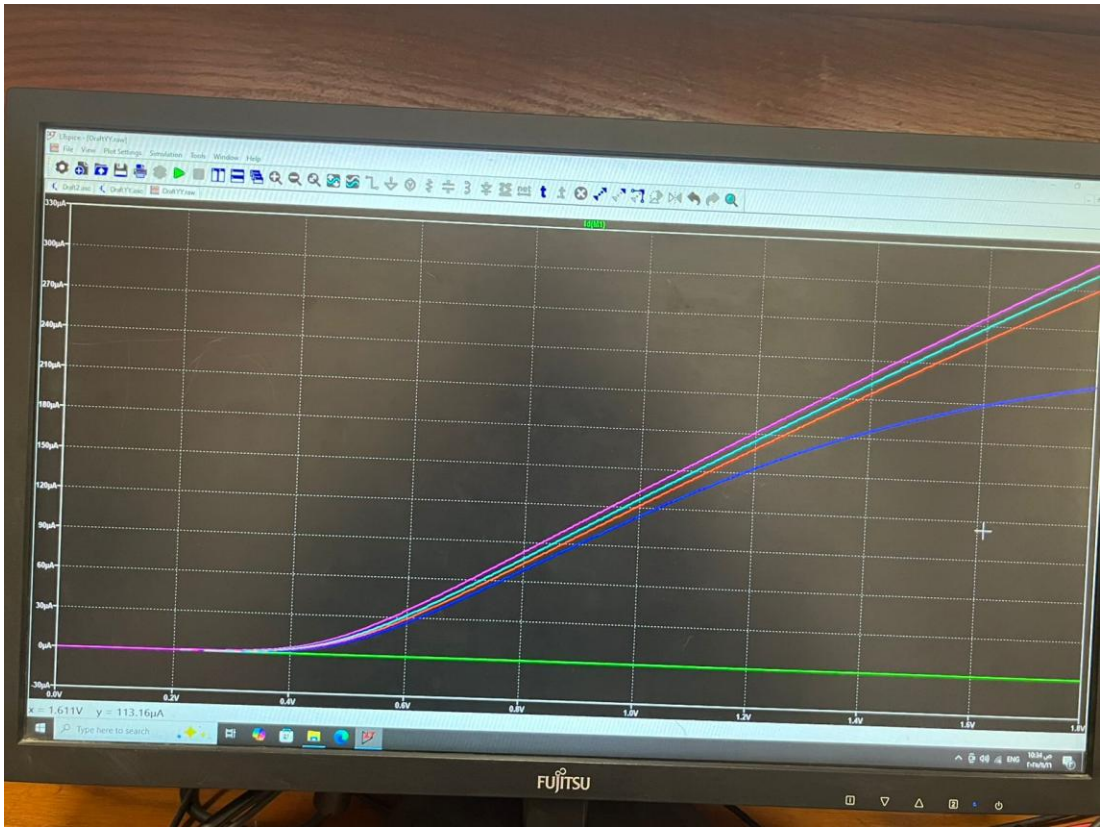
Part 1: MOSFET Characteristics

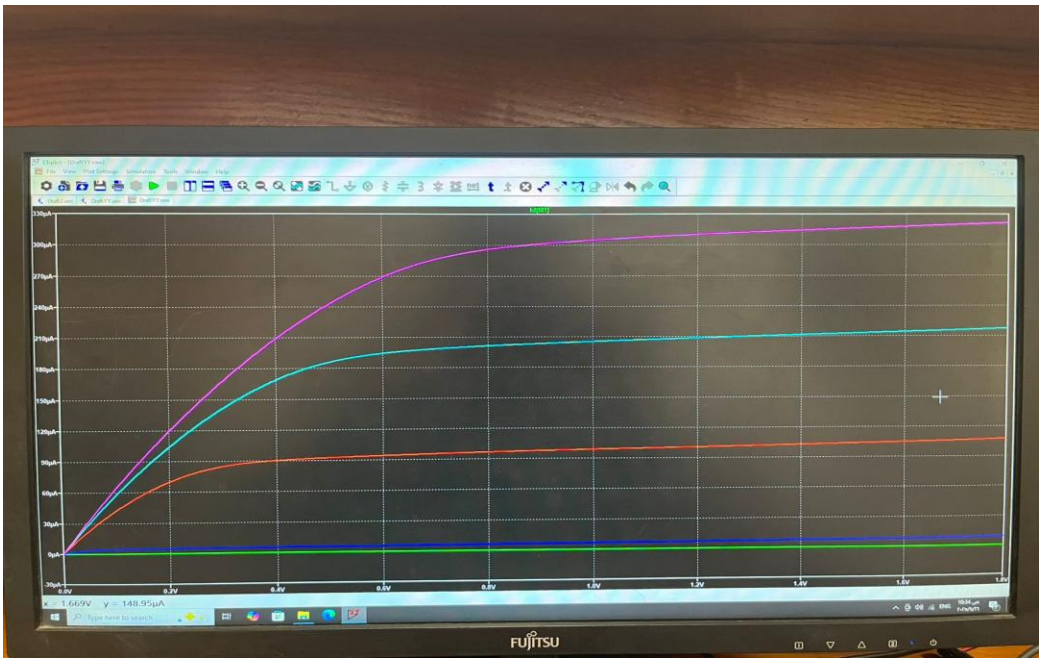




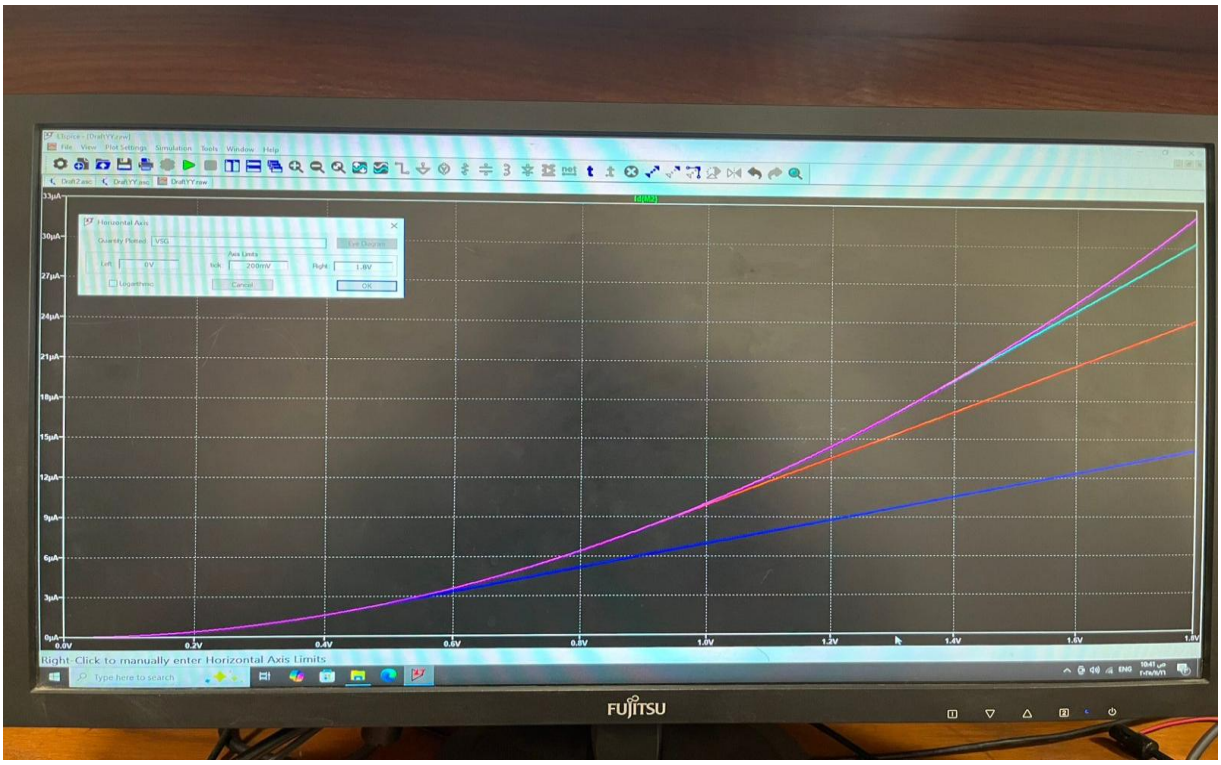
- **Screenshots of the IDS vs VDS, IDS vs VGS graphs for NMOS, and PMOS transistors.**

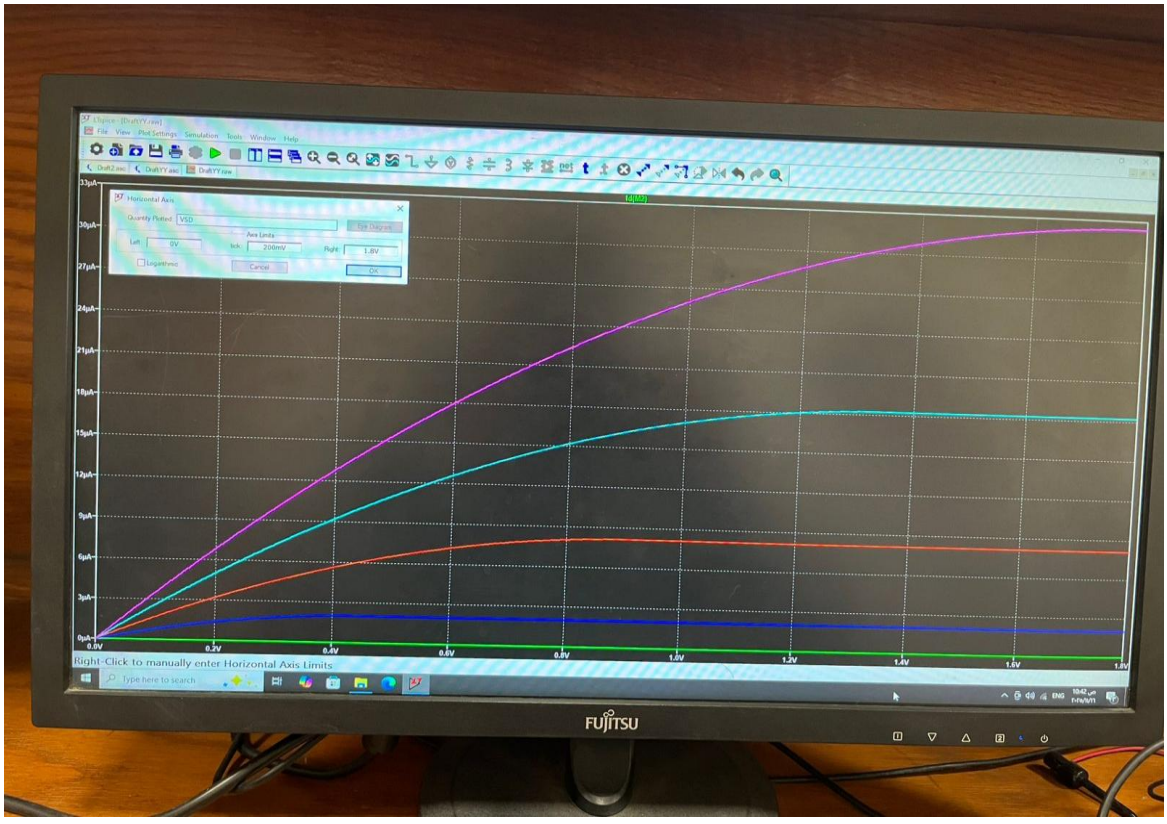
NMOS :



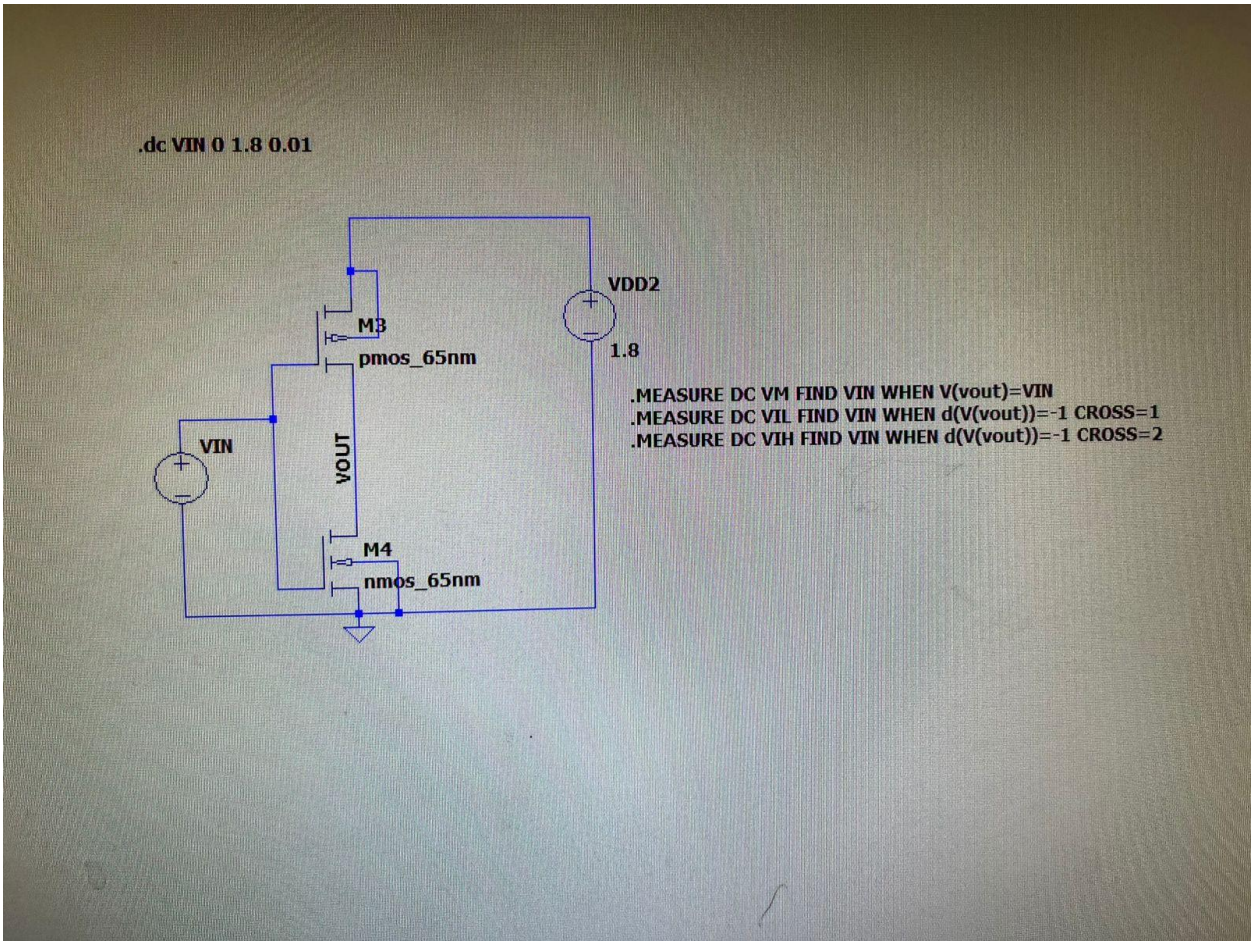


PMOS :

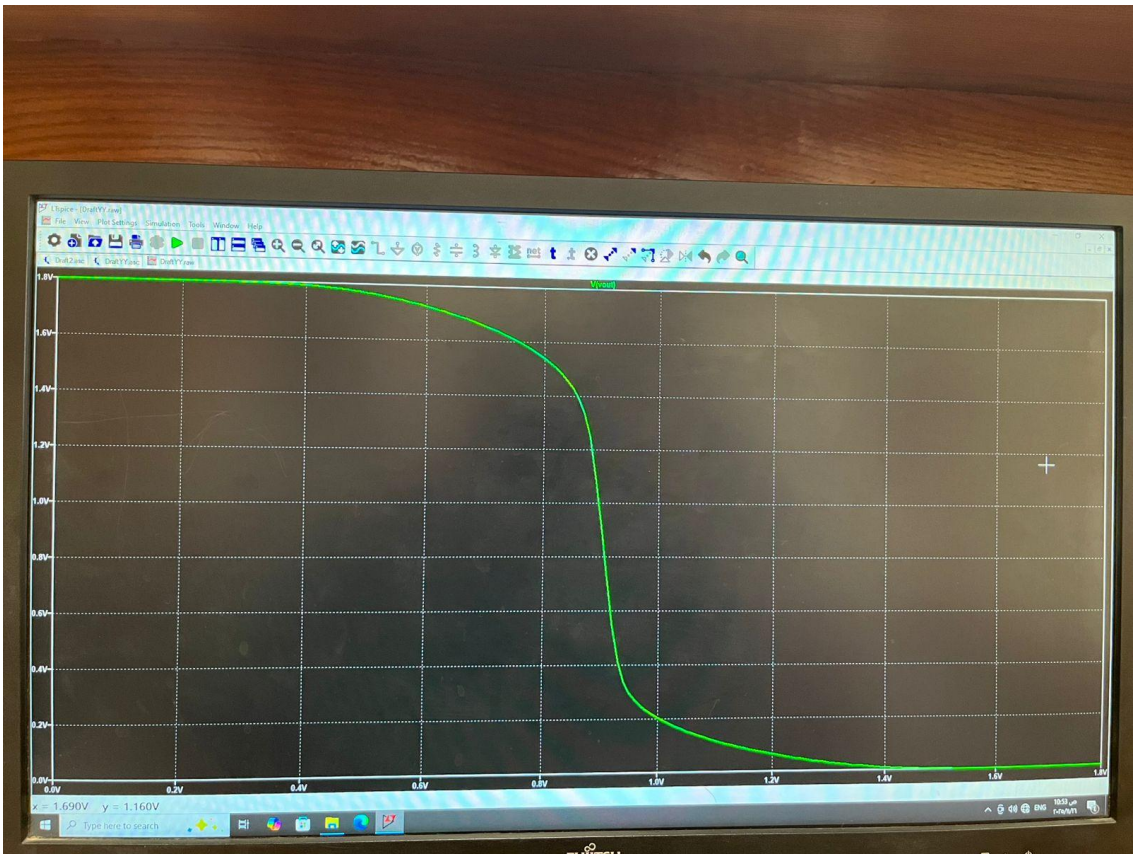

Right-Click to manually enter Horizontal Axis Limits
FUJITSU


Right-Click to manually enter Horizontal Axis Limits
FUJITSU

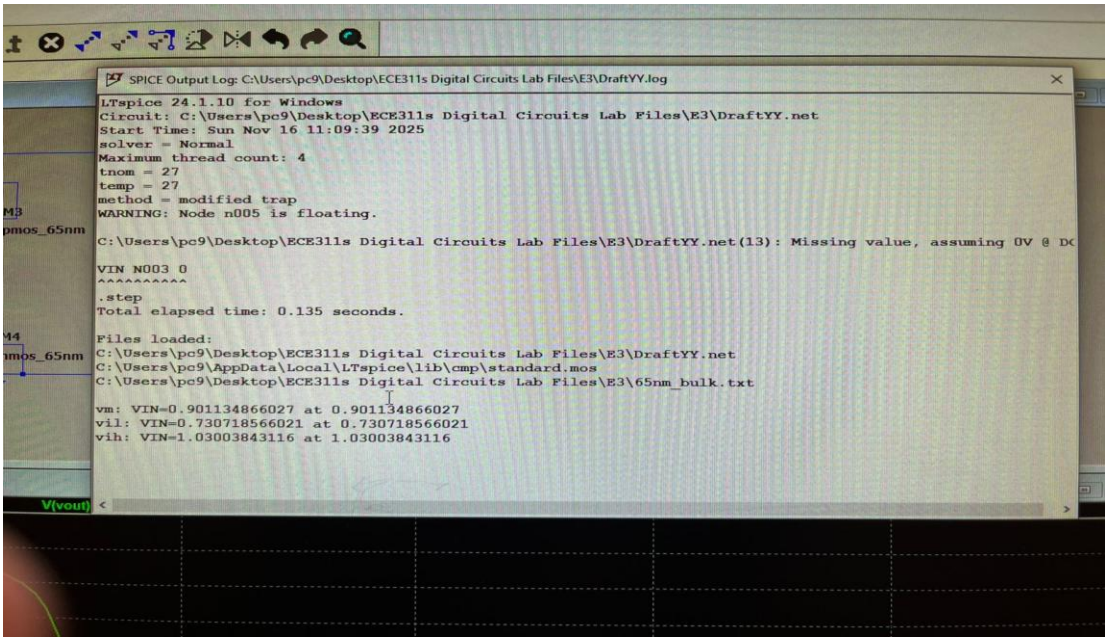
Part 2: CMOS Inverter VTC Simulation



- **From DC sweep, screenshot of CMOS inverter VTC.**



- **Measure the values of VM, VOH, VOL, VIH, and VIL hence calculate the Noise margins of the CMOS inverter**



VM = 0.9 V

VIL = 0.73 V

VIH = 1.03 V

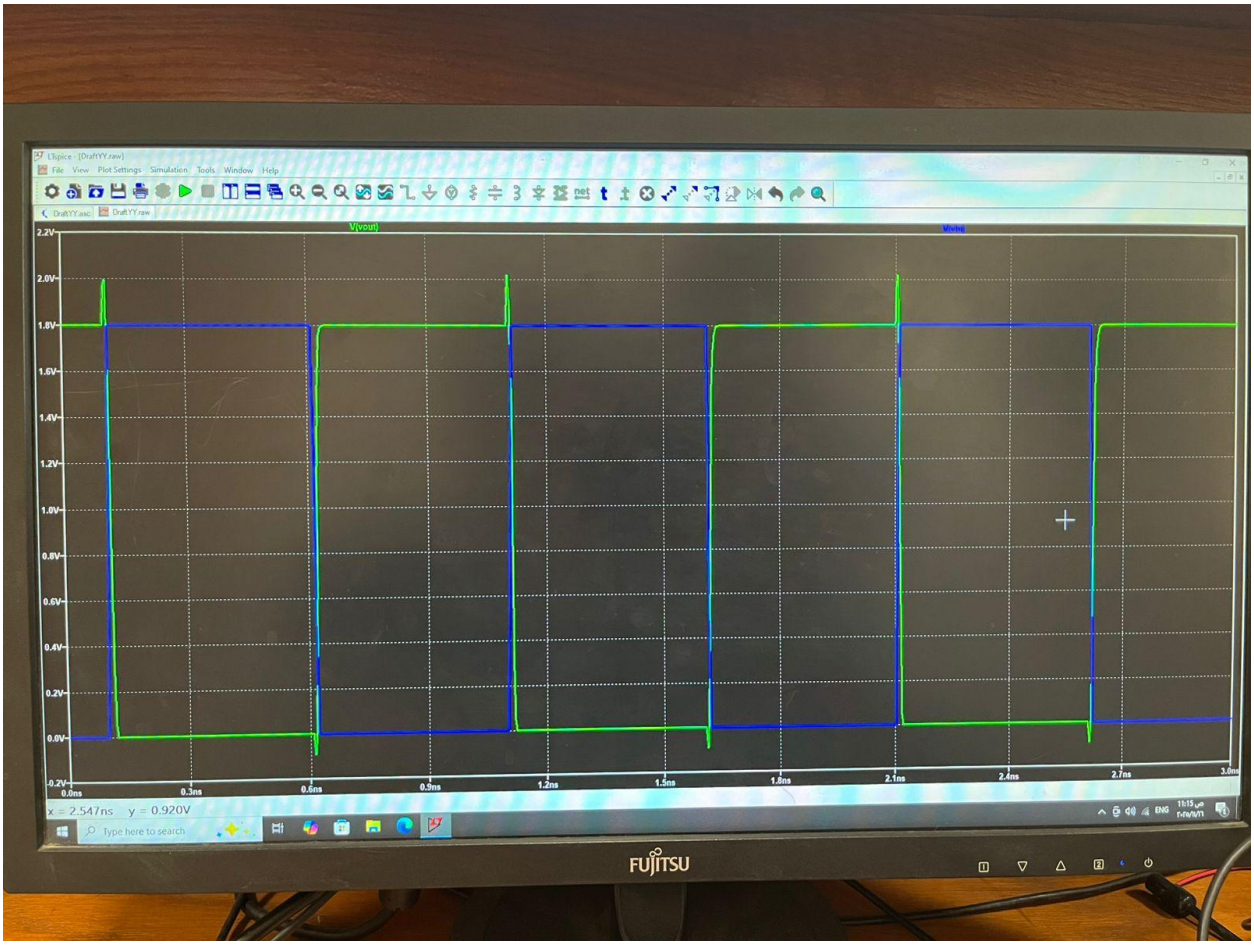
VOH = 1.8 V

VOL = 0 V

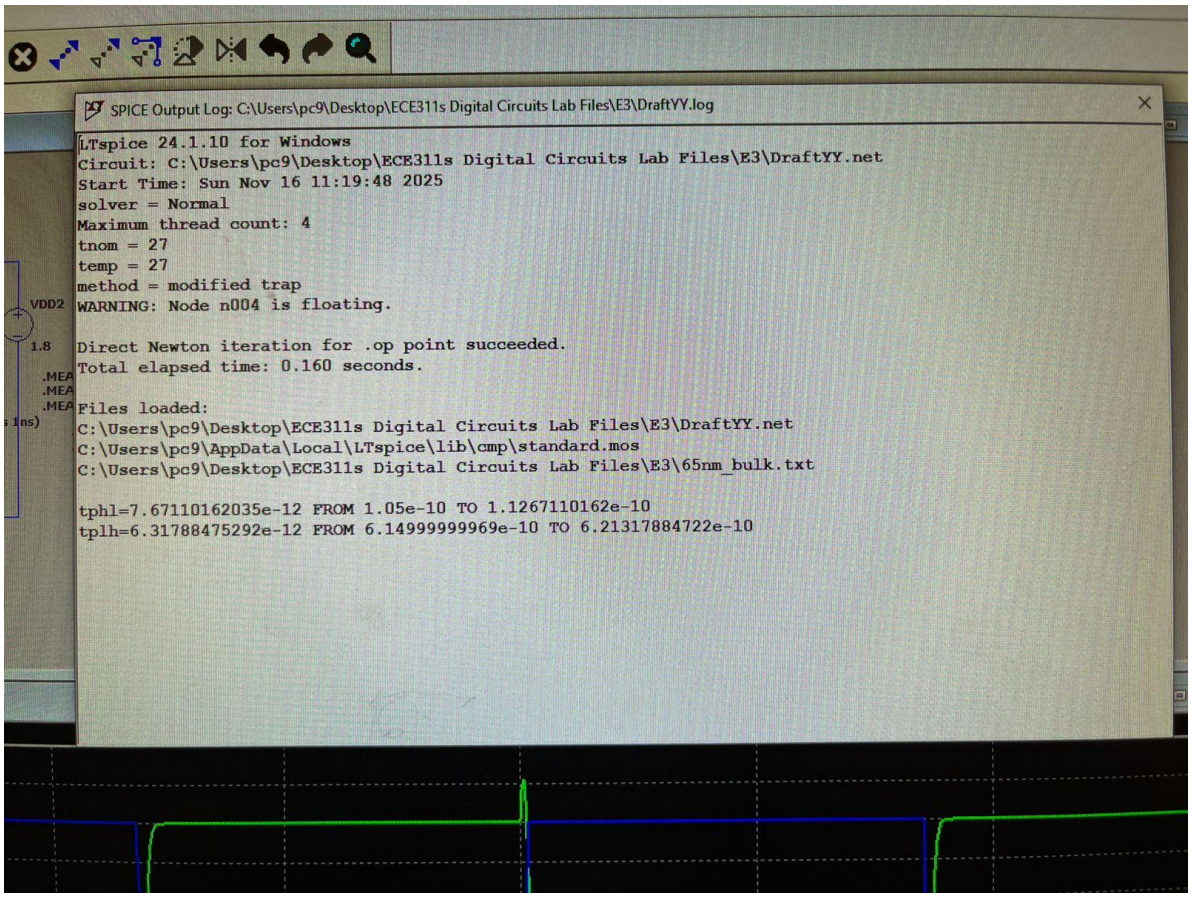
NMH = VOH – VIH = 1.8 – 1.03 = 0.77 V

NML = VIL – VOL = 0.73 – 0 = 0.73 V

- **From Transient simulation, screenshot showing both Vin and Vout on same plot.**



- **Measure the propagation delays from the plot TpLH and TpHL , hence calculate Tp.**

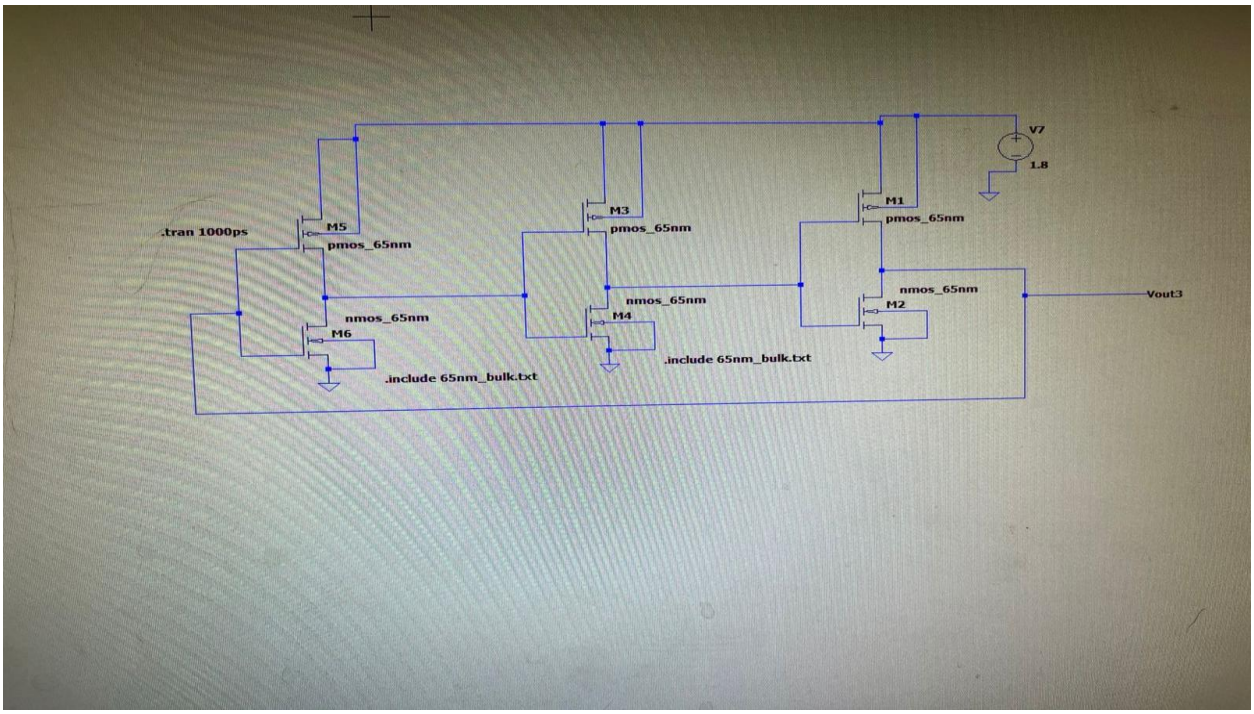


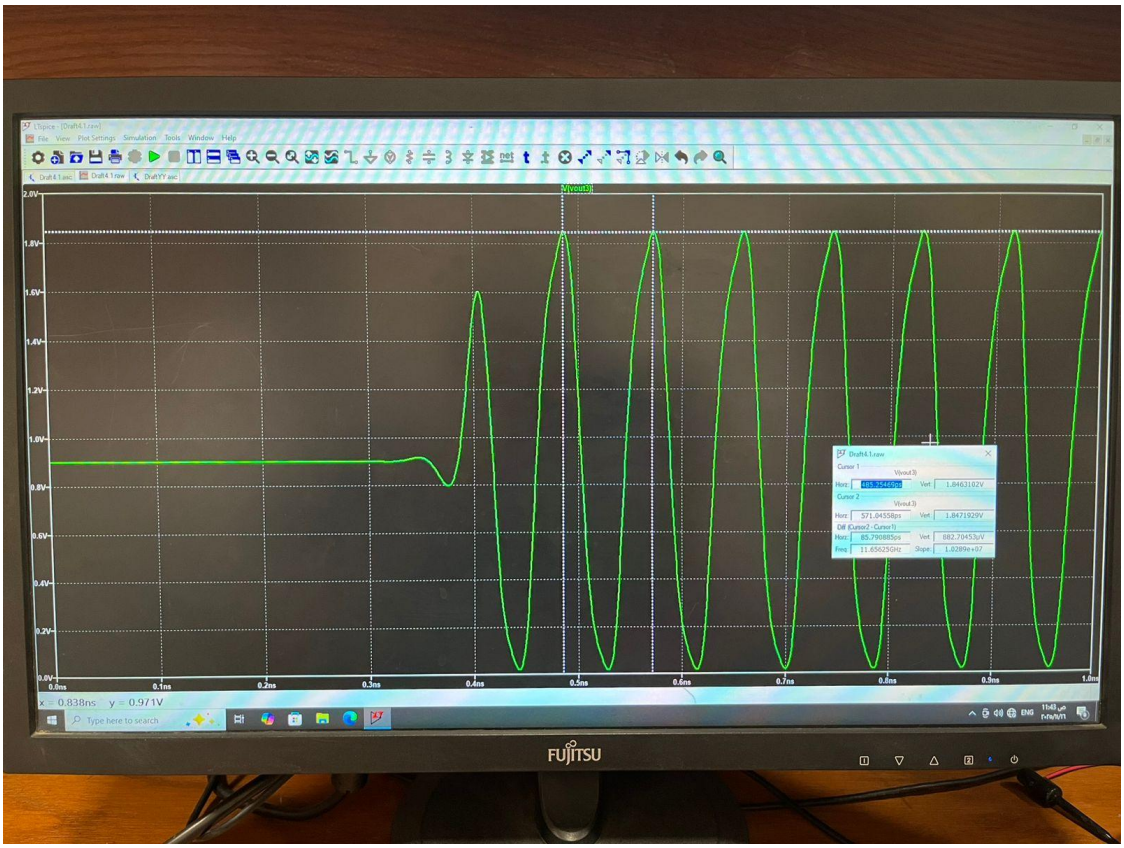
Tphl = 7.7x10power-12

Tplh = 6.3x10power-12

Tp = (tphl + tplh) / 2 = 7x10power-12

Part 3: Three-Stage Ring Oscillator





The value of the measured oscillation frequency = 11.66 GHz